

Exploration of in-situ utilization of extraterrestrial resources: Preparation of Chang'E-5 lunar regolith simulant geopolymers modified by Al_2O_3 and FeTiO_3 under vacuum environment

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ABSTRACT

Geopolymer solidification is considered a promising strategy for in situ resource utilization (ISRU) in lunar base construction through the synthesis of building materials from lunar regolith. In this study, a Chang'E-5 lunar regolith simulant (CE-5 LRS) named DH-1, characterized by low aluminum weight percent (Al_2O_3 , 11.2 wt%) and high ilmenite content (FeTiO_3 , 6.0 wt%), was used to prepare geopolymers under simulated lunar conditions. The effects of curing temperature, atmosphere, and additives ($\gamma\text{-Al}_2\text{O}_3$ and FeTiO_3) were systematically investigated, revealing notable changes in microstructure and mechanical properties. Thermal analysis demonstrated that higher average curing temperatures ($36.5\text{--}76.2\text{ }^\circ\text{C}$) improved mechanical performance, with the optimal specimen achieving a compressive strength of $12.0 \pm 0.5\text{ MPa}$. Vacuum-cured specimens exhibited 62.7 % increase in strength compared to air-controls, due to the absence of CO_2 , which prevented incomplete reactions with the NaOH activator. The addition of $\gamma\text{-Al}_2\text{O}_3$ led to a non-linear effect on strength. Incorporating 5 wt% $\gamma\text{-Al}_2\text{O}_3$ increased strength by 36.5 % over the baseline due to enhanced formation of crosslinked binding gels. However, higher amounts (10–15 wt%) caused a decline in strength owing to microcracking and the formation of amorphous $\text{Al}(\text{OH})_3$. In contrast, FeTiO_3 additions (5–15 wt%) displayed negligible influence on the reaction process and final strength, indicating its inert behavior under alkaline conditions.

1. Introduction

The Moon has emerged as a strategic extraterrestrial platform for advancing humanity's deep space exploration initiatives. As Earth's closest celestial body, it offers unique scientific opportunities, accessible resources, and a vital location for future interplanetary missions [1,2]. These advantages support the goal of building a permanent lunar base. Such construction will require large amounts of structural and functional materials. However, terrestrial material transport remains prohibitively expensive, with NASA estimating lunar orbit delivery costs at approximately 3370 \$/kg [3]. This makes it essential to use local resources, especially lunar regolith (LR), a complex surface material that requires a multidisciplinary approach combining materials engineering, manufacturing systems, and structural design [4,5].

Researchers around the world have proposed various techniques for the preparation of building materials from LRS, which is a replica of Earth minerals with physical and chemical properties similar to those of real lunar regolith [6–8]. These include melt casting [9,10], 3D printing [11,12], dry consolidation [13], geopolymerization [14–18], and so on. Among them, geopolymerization is regarded as a promising technique, requiring only simple molds for preparation and can react at low temperatures [19,20]. The geopolymerization reaction yields an amorphous to semi-crystalline aluminosilicate gel matrix that serves as the primary binding phase. In sodium-activated systems, this gel is conventionally designated as N-A-S-H (where $\text{N} = \text{Na}_2\text{O}$, $\text{A} = \text{Al}_2\text{O}_3$, $\text{S} = \text{SiO}_2$, $\text{H} = \text{H}_2\text{O}$), which plays a critical role in determining the mechanical performance of the resulting material [21]. Given the potential presence of water and ice on the Moon, all essential raw materials required for producing

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geopolymer [22].

To produce high-performance geopolymers suited to the extreme lunar environment, researchers have explored different factors. Montes [23] et al. prepared the JSC-1A LRS based geopolymers. It was found that the resulting geopolymer possesses good radiation shielding capabilities. Zhou [22] et al. studied the effects of sodium hydroxide (m_{SH}) and sodium silicate (m_{SS}) on the mechanical properties and microstructure of the geopolymer samples. The geopolymer exhibited the highest compressive strength and flexural strength when the m_{SH} / m_{SS} ratio was 0.6. Zhang [24] et al. synthesized BH-1 LRS based geopolymers and treated them under low and high temperature environments to investigate the durability of the geopolymer. The results showed that the mechanical properties of the geopolymers underwent significant degradation, with flexural strength decreasing by approximately 40 %. This was attributed to granulation of the geopolymer structure, accompanied by visible cracks and increased porosity. Xue [25] et al. proposed a calculation method coupling thermodynamic modeling with an experimental study to elucidate the evolutionary relationship between the pore solution, product formation, and compressive strength. Their findings demonstrated that Si and Al concentrations in the pore solution, as well as gel product formation, are key factors affecting compressive strength.

However, previous works have primarily focused on Apollo-series LRS with similar chemical and mineral compositions [26]. A significant research gap exists regarding geopolymers derived from the CE-5 LR, which notably exhibits the lowest Al_2O_3 content and elevated $FeTiO_3$ concentration among all LR samples [27–29]. From an ISRU perspective, the molar ratio of Si and Al in the LR critically influences

the strength development of geopolymers. Given the low Al_2O_3 content in CE-5 LR, it is essential to determine the required extent of Al supplementation and identify the optimal incorporation mechanism to enable effective fabrication of construction materials from local resources. Furthermore, the high $FeTiO_3$ content in CE-5 LR is significant for both geological and ISRU reasons. Mineralogically, it distinguishes the young basalts sampled by Chang'E-5 from older lunar terrains. Moreover, $FeTiO_3$ is a potential ISRU resource for oxygen and metal production. Determining its behavior during geopolymerization, as a reactive component or inert filler, is thus essential for assessing the feasibility of using LR in construction. Additionally, the systemic understanding of how the Moon's high-vacuum environment and extreme temperature variations influence the geopolymerization reaction is still lacking [30,31].

To this end, the feasibility of utilizing DH-1 under a simulated lunar temperature and vacuum was verified. Comparative air-cured specimens were prepared to isolate and investigate the vacuum-induced changes. The optimal $\gamma-Al_2O_3$ addition for compensating the inherent low Al content was determined, and its progressive incorporation effect on the microstructural evolution was revealed. In addition, the role of the abundant $FeTiO_3$ was conclusively identified, confirming its chemical inertness and defining its sole contribution through physical densification mechanisms. This work provides fundamental insights and a material design strategy for the ISRU of LRS.

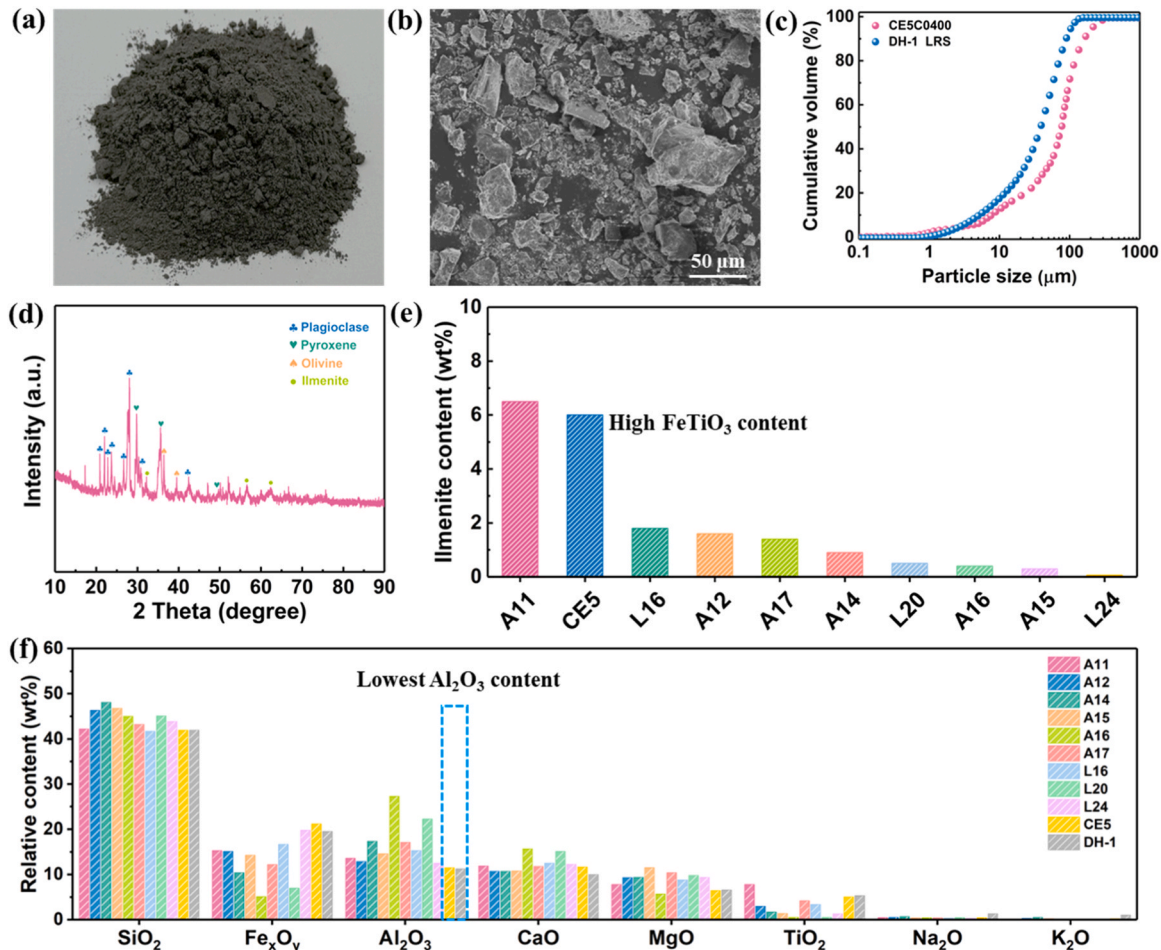


Fig. 1. (a) Optical picture, (b) SEM image, (c) Particle size distribution and (d) Mineral properties of the DH-1. (e) $FeTiO_3$ content of several lunar regolith samples. (f) Chemical composition of the DH-1 and several lunar regolith samples.

2. Experimental

2.1. Raw materials

The DH-1 LRS was developed in-house by drying, grinding, sieving of natural minerals and then mixing with a specific proportion of FeTiO_3 powder. As shown in Fig. 1a (optical image) and Fig. 1b (SEM image), DH-1 is gray in color and consists of angular particles. Comparative particle size analysis (Fig. 1c) confirms a similar particle size distribution between DH-1 and CE-5 LR (CE5C0400) [27]. XRD pattern (Fig. 1d) identifies main mineral composition of plagioclase, pyroxene, olivine, and ilmenite of DH-1, aligning with CE-5 LR mineralogy. Fig. 1e reveals CE-5 samples contain elevated FeTiO_3 relative to Apollo and Luna missions, except Apollo11. Chemical composition analysis via XRF (Fig. 1f) demonstrates DH-1 successfully replicates CE-5 LR geochemistry, with lower Al_2O_3 levels than Apollo and luna samples (11.2 vs. 12.5–27.3 wt%) [29].

2.2. Geopolymer preparation

Fig. 2 presents the fabrication process of the DH-1 based geopolymers. The alkaline activator was prepared by dissolving anhydrous NaOH pellets ($\geq 96\%$ purity, Sinopharm Chemical Reagent Co., Ltd., China) in deionized water, followed by 24 h maturation at room temperature. Through trial-and-error testing, the optimal NaOH/DH-1 mass ratio was established at 12 wt% in the control group (V-II). On this basis, in light of inherent low Al_2O_3 and elevated FeTiO_3 concentration in DH-1, stoichiometric additions of $\gamma\text{-Al}_2\text{O}_3$ (analytical purity, Sinopharm Chemical Reagent Co., Ltd., China) and FeTiO_3 (99.8 % purity, Zishu Mineral Products Co., Ltd., China) were incorporated at 5, 10, and 15 wt % relative to the DH-1 mass. The dosage levels of $\gamma\text{-Al}_2\text{O}_3$ (5, 10, 15 wt %) were selected based on stoichiometric considerations. The 5 wt% raises the total Al_2O_3 content to ~ 16 wt%, within the typical range for effective geopolymerization, while higher doses (10–15 wt%) were used to establish a dose-response relationship and examine the effects of excess Al. FeTiO_3 was added at the same levels (5–15 wt%) to simulate ilmenite-rich LR and systematically study its influence as an inert filler. The precursor mixture, with a water-to-solid ratio of 0.28, was

homogenized using a planetary mixer: primary mixing at 150 ± 5 rpm for 120 s followed by high-speed conditioning at 300 ± 10 rpm for an additional 120 s. The evenly mixed slurry was cast into steel molds in the dimension of $50 \text{ mm} \times 50 \text{ mm} \times 50 \text{ mm}$ compacted by a vibration table for 30 s. Molds were subsequently covered with polyethylene film and cured under different temperature and atmospheric conditions. The chamber was evacuated to a gauge pressure of -0.1 MPa (absolute pressure of $\sim 0.01 \text{ MPa}$), effectively removing the atmospheric gases (including N_2 , O_2 , CO_2 , and water vapor) to create a low-pressure, dry lunar environment. CO_2 concentration and vacuum level were continuously monitored using a resistance vacuum gauge to ensure environmental stability. The specific mix design of the DH-1 based geopolymers is listed Table 1.

The curing temperature for the DH-1 based geopolymers was determined by the real ambient temperature recorded by the Diviner Lunar Radiometer Experiment onboard the Lunar Reconnaissance Orbiter (LRO) [32–34]. As indicated in Fig. 3a, the CE-5 landing site is near 45° latitude, so the temperature profile at this latitude was adopted. A lunar day is conventionally considered equivalent to 30 Earth days. Therefore, the Diviner temperature data (Fig. 3b) were scaled by a factor of 30 to represent surface temperature variations over 720 h (Earth time). Earth time was used in this study to make the statement of time

Table 1
Mix design of the DH-1 based geopolymers.

Test No.	$m_{\gamma\text{-Al}_2\text{O}_3}/m_{\text{DH-1}}$	$m_{\text{FeTiO}_3}/m_{\text{DH-1}}$	Scheme	Atmosphere
V-I	-	-	I	Vacuum
V-II	-	-	II	Vacuum
V-III	-	-	III	Vacuum
V-IV	-	-	IV	Vacuum
V-V	-	-	V	Vacuum
A-II	-	-	II	Air
V-A5	5 %	-	II	Vacuum
V-A10	10 %	-	II	Vacuum
V-A15	15 %	-	II	Vacuum
V-F5	-	5 %	II	Vacuum
V-F10	-	10 %	II	Vacuum
V-F15	-	15 %	II	Vacuum

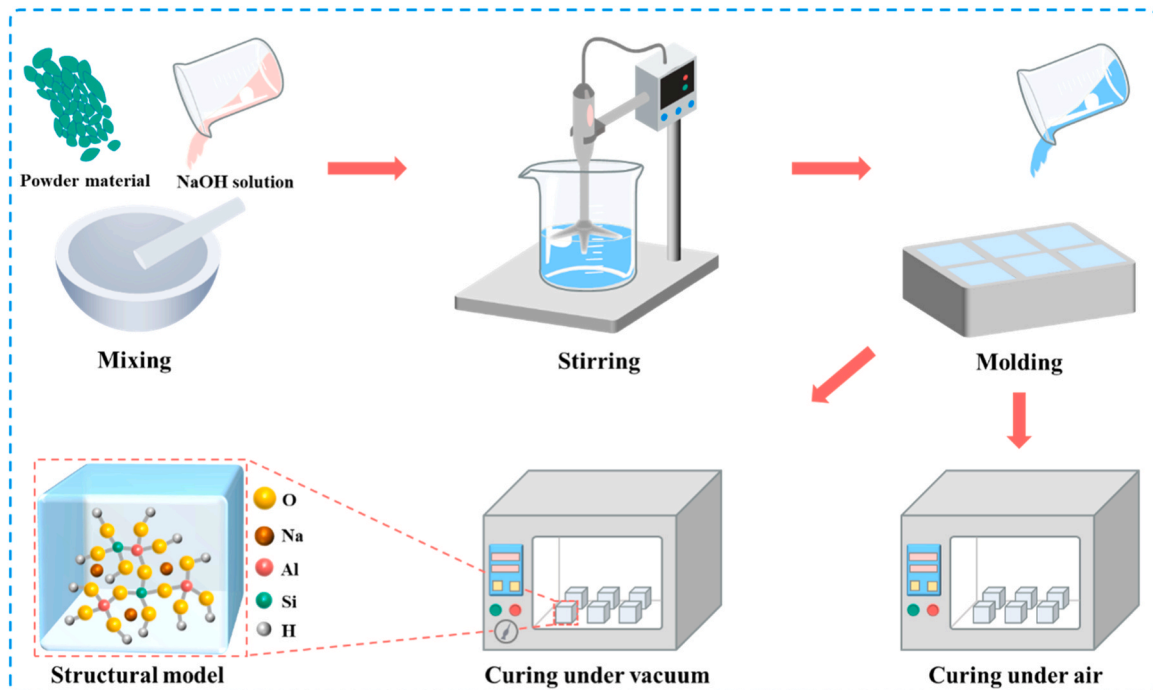


Fig. 2. Schematic diagram of the preparation of the DH-1 based geopolymers.

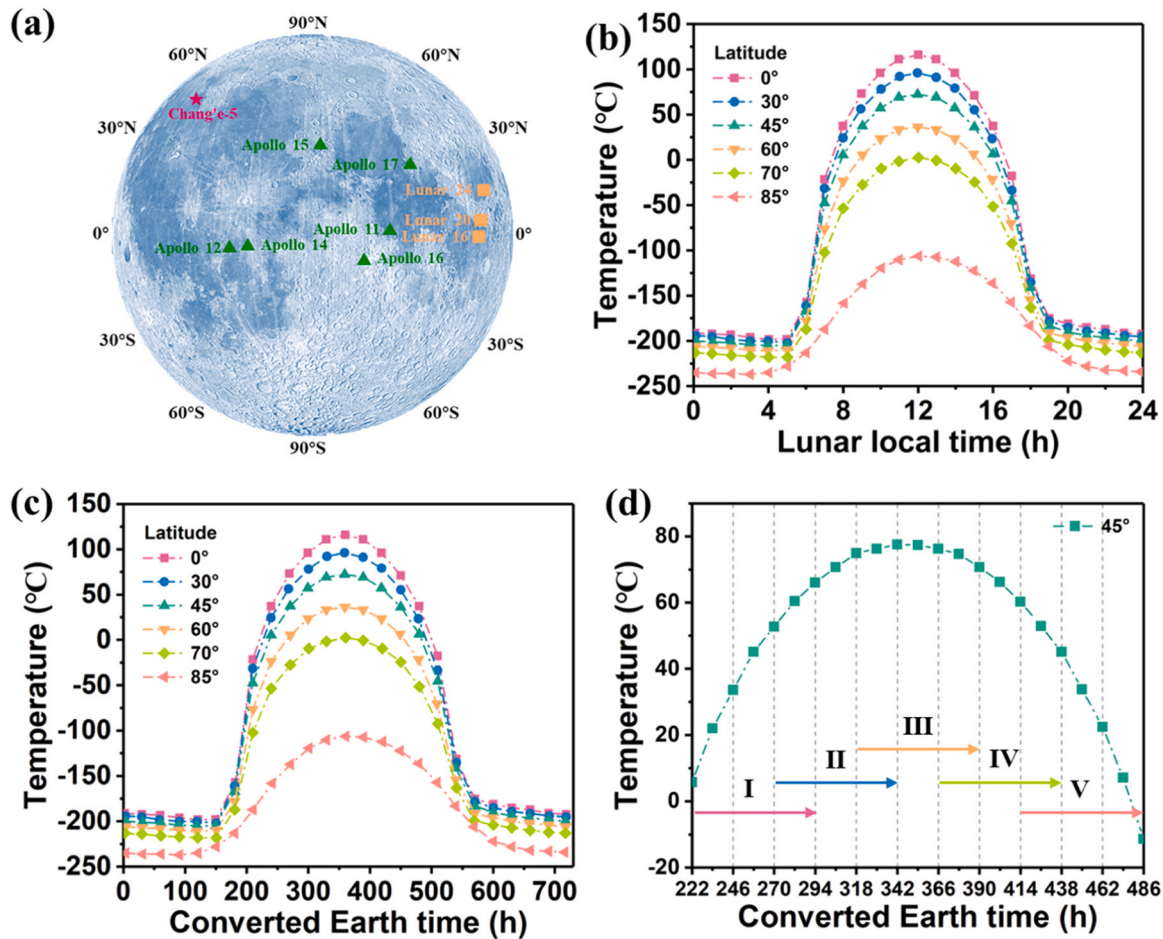


Fig. 3. (a) The landing positions of spacecraft on the Moon. (b) Lunar surface temperature. (c) Converted Earth time. (d) Curing temperature extracted by linear interpolation method.

clearer. After conversion, the time interval between two adjacent data points was 30 h (Fig. 3c). Linear interpolation was applied to refine the data to 12-hour intervals. Only temperatures above 0°C were considered. The intervals labeled I–V correspond to 222–294 h, 270–342 h, 318–390 h, 366–438 h, and 414–486 h, respectively (Fig. 3d). Linear interpolation is a widely accepted method for reconstructing continuous signals from discrete measurements and has been used in previous studies [32,34]. The curing profile was defined with a temporal resolution of 12-hour. This interval was chosen to balance the accurate representation of major thermal trends on the lunar surface with practical experimental controllability over the 72-hour curing process. Although finer intervals could resolve more detailed temperature variations, the 12-hour resolution adequately captures the key heating and cooling phases that predominantly influence geopolymerization kinetics. Most importantly, it effectively represents the temperature differences among various schemes during the critical early stage (0–24 h), which our findings establish as the decisive factor affecting the final mechanical properties (see Section 3.1). The curing temperature schemes simulating lunar surface conditions at 45° latitude are detailed in Table 2.

2.3. Testing methods

Uniaxial compressive strength of the DH-1 based geopolymers were test after 3 days and 28 days of curing according to ASTM C109 [35]. Three valid measurements were taken, and the average was recorded as the final value. The loading speed for the compressive strength tests was set to 0.1 mm/min. Fig. 4a and b present part of the specimens and the

Table 2

Curing temperature schemes from I to V derived by real lunar surface temperature.

Curing time	Group of curing temperature scheme				
	I (°C)	II (°C)	III (°C)	IV (°C)	V (°C)
0–12 h	5.9	52.7	74.9	76.3	60.3
12–24 h	21.8	60.2	76.2	74.9	52.7
24–36 h	33.5	66.1	77.5	70.6	44.9
36–48 h	45.1	70.5	77.4	66.3	33.7
48–60 h	52.7	74.9	76.3	60.3	22.5
60–72 h	60.2	76.2	74.9	52.7	7.4
Average (24 h)	13.9	56.4	75.5	75.6	56.5
Average (72 h)	36.5	66.8	76.2	66.9	36.9

setup for the compression test.

X-ray diffraction (XRD, D8, Bruker, Germany) were conducted in the range of 5–90° (2θ) with a scanning rate of 2°/min to analyze the mineral phases in DH-1 and reaction products.

Field emission scanning electron microscopy (FESEM, 15 kV, S4800, Hitachi, Japan) coupled with energy dispersive spectrometry (EDS) was employed for micromorphology and elemental components observation.

Fourier infrared spectroscopy (FTIR, NEXUS-670, Thermo Nicolet, USA) was performed to identify the structural changes in functional amorphous phase in wavenumber from 400 cm⁻¹ to 4000 cm⁻¹.

²⁷Al magic angle spinning nuclear magnetic resonance spectroscopy (MAS NMR, 600 M, Agilent, USA) solid-state spectra were recorded to determine Al coordination in V-II and Al-added geopolymers.

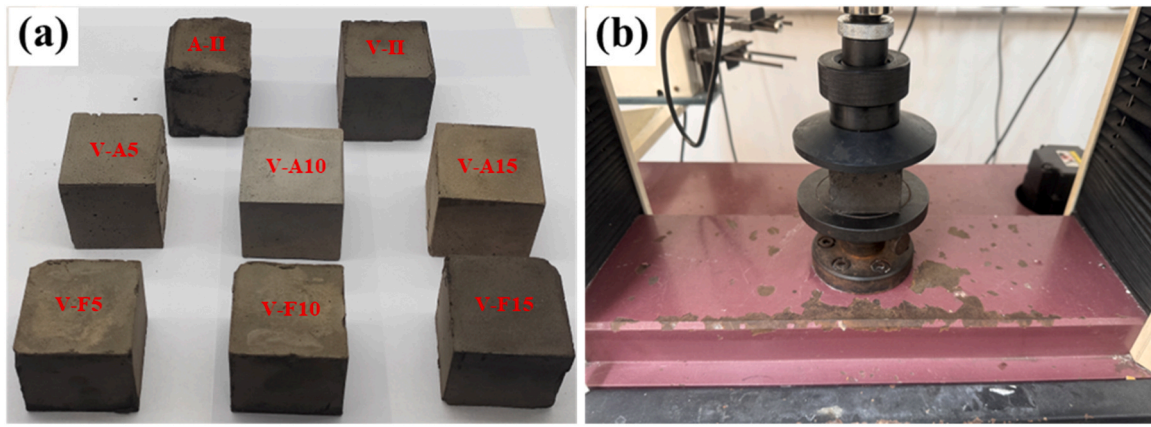


Fig. 4. (a) optical picture and (b) compressive strength test set of the DH-1 based geopolymers.

3. Results and discussion

3.1. Compressive strength results

The 3-day compressive strength test results of the DH-1 based geopolymers are shown in Fig. 5. The specimens exhibit significant strength variations ranging from 2.4 ± 0.2 MPa (V-I) to 12.0 ± 0.5 MPa (V-III), representing a fivefold difference in mechanical performance (Fig. 5a). Curing temperature schemes (Table 2) reveal distinct curing kinetics, with scheme III maintaining elevated average temperatures (75.5°C at 24 h; 76.2°C at 72 h) compared to scheme I (13.9°C and 36.5°C , respectively), confirming the positive correlation between thermal energy input and geopolymerization efficiency. Notably, scheme II and IV

demonstrate comparable average temperature (66.8°C vs. 66.9°C at 72 h), yet V-IV achieves superior strength (10.2 ± 0.4 MPa vs. 9.6 ± 0.3 MPa). This difference originates from their various early-stage curing temperature, the higher initial temperature (75.6°C vs. 56.4°C at 24 h) of scheme IV facilitates rapid geopolymer network formation prior to water dissipation. Similarly, scheme I and V share equivalent 72-h average temperature (36.5°C and 36.9°C), but the enhanced early thermal activation of V-V (56.5°C vs. 13.9°C at 24 h) yields 121 % strength improvement (5.3 ± 0.9 MPa vs. 2.4 ± 0.2 MPa).

The Vacuum-cured specimen V-II shows a 62.7 % higher compressive strength (9.6 ± 0.3 MPa) than its air-cured counterpart A-II (5.9 ± 0.8 MPa) (Fig. 5b). This vacuum-induced strengthening effect aligns with lunar environmental conditions, suggesting extraterrestrial curing

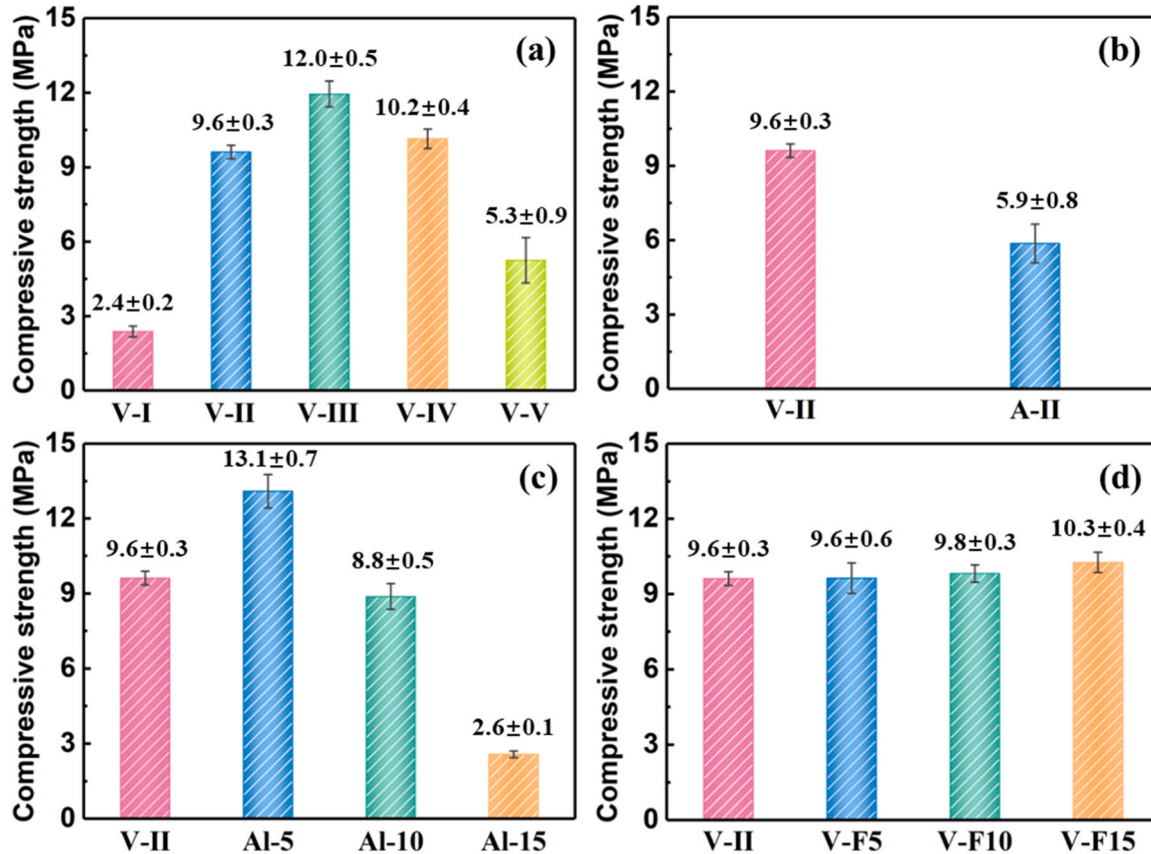


Fig. 5. The 3-day compressive strength of the DH-1 based geopolymers. (a) I-V group, (b) V-II and A-II, (c) V-II and $\gamma\text{-Al}_2\text{O}_3$ additives group, and (d) V-II and FeTiO_3 additives group.

environments may optimize the mechanical performance of LR based geopolymers.

Fig. 5c illustrates the effect of γ -Al₂O₃ addition on mechanical behavior, implying a non-monotonic relationship. Specimen V-A5 (5 wt % γ -Al₂O₃ addition) achieves the highest compressive strength of 13.1 ± 0.7 MPa, representing a 36.5 % enhancement over the V-II baseline (9.6 ± 0.3 MPa). Conversely, V-A15 (15 wt% γ -Al₂O₃ addition) exhibits severe strength reduction to 2.6 ± 0.1 MPa. This suggests that optimal mechanical properties occur at intermediate γ -Al₂O₃ additions (5–10 wt %), where balanced charge compensation and gel network connectivity are achieved. Fig. 5d presents the limited effect of FeTiO₃ additives, with strength increasing only 7.3 % (9.6 ± 0.3 – 10.3 ± 0.4 MPa) across the tested range.

The 28-day compressive strength of the DH-1 based geopolymers is evaluated to assess their long-term mechanical performance, as depicted in Fig. S1. All specimens exhibit strength enhancements after 28 days of curing compared to the 3-day results, with strength values increasing from 3.0 ± 0.2 MPa (V-I) to 13.7 ± 0.5 MPa (V-III), corresponding to a relative improvement of 14.1–25.0 %. Specimens cured under higher average temperatures display diminished smaller gains. Under identical curing scheme II, vacuum-cured and air-cured specimens reach compressive strengths of 11.1 ± 0.6 MPa and 6.9 ± 0.4 MPa, increases of 15.6 % and 16.9 %, respectively. Adding 5 wt% and 10 wt% γ -Al₂O₃ resulted in strength increases of 12.2 % (14.7 ± 0.7 MPa for V-A5) and 13.6 % (10.0 ± 0.4 MPa for V-A10), respectively. In contrast, the specimen with 15 wt% γ -Al₂O₃ addition shows a significant strength reduction of 53.8 % (1.2 ± 0.1 MPa for V-A15), making it the only formulation with mechanical degradation. In addition, FeTiO₃-added specimens demonstrate consistent strength improvement of about 16.0 %, achieving final strength values between 11.2 ± 0.4 MPa (V-F5) and 11.9 ± 0.4 MPa (V-F15).

3.2. XRD analysis

XRD analysis (Fig. 6a) confirms the crystalline signatures of supplementary materials, with γ -Al₂O₃ exhibiting characteristic peaks at 37.3° , 45.6° , and 67.2° , while FeTiO₃ displays predominant peaks at 32.5° , 35.4° , and 54.3° . The geopolymers (Fig. 6b) largely retained the positions and intensities of raw DH-1 peaks, indicating that the main reaction products were amorphous. However, two distinct crystalline peaks emerge at 14.0° and 24.4° , which are attributed to the formation of zeolitic phases [36,37]. Their relative intensities are shown in Fig. S2.

Notably, the peak intensities in specimen V-II are approximately 10 % higher than in A-II. In addition, γ -Al₂O₃ addition (5–15 wt%) had a non-linear effect on peak intensity. Specifically, specimen V-A10 demonstrates the highest intensities (1.2 times greater than V-II), while V-A15 shows a 30 % reduction relative to V-A10. In contrast, the introduction of FeTiO₃ has no significant effect on the peak intensities, which remain largely unchanged compared to specimen V-II. Complete dissolution of γ -Al₂O₃ signatures confirm effective alkaline activation, whereas persistent FeTiO₃ reflections at 32.2° demonstrates chemical inertness in NaOH solution. Note that moderate crystallized phase can fill into the cavity of the three-dimensional network structure of geopolymer making the structure denser combined with the amorphous products [38].

3.3. SEM and EDS analysis

Microstructural evolution of the DH-1 based geopolymers is characterized through SEM (Fig. 7). The air-cured specimen A-II (Fig. 7a) exhibits several unreacted DH-1 particles interlaced with the amorphous gel generated by geopolymerization. Higher magnification (Fig. 7b) displays the initial growth of gel on the particle surface, indicating an insufficient reaction. The vacuum-cured specimen V-II (Fig. 7c) demonstrates enhanced geopolymerization kinetics and better three-dimensional gel connectivity. However, there are still recognizable particles that are not fully encapsulated by the amorphous gel.

Satisfying microstructural integration is achieved in specimen V-A5 (Fig. 7d), all DH-1 particles are encapsulated in amorphous gel, forming a nearly complete whole. V-A10 presents the most formation of gel, however, obvious cracks can be observed in Fig. 7e. In contrast, the gel in V-A15 decreases dramatically compared to V-A5 (Fig. 7f). FeTiO₃ added geopolymers (Fig. 7g-i) exhibit similar micromorphology to V-II, with no significant enhancement in gel formation. However, several unreacted particles are observed embedded within the gel matrix, particularly in specimen V-F15 with the highest addition level (Fig. 7i). EDS spot analysis (Fig. S3) confirms these were FeTiO₃ particles, rich in O, Fe, and Ti.

High-resolution SEM analysis (Fig. 8) reveals distinct non-gel product evolution. A-II (Fig. 8a) exhibits almost negligible product, while V-II (Fig. 8b) develops short acicular product. Increasing γ -Al₂O₃ addition (Fig. 8c-f) induces systematic enhancement in both abundance and dimension of the product, with the V-A15 is full of flaky non-gel product. EDS spot results (Fig. S4) reveal that the O and Al dominated

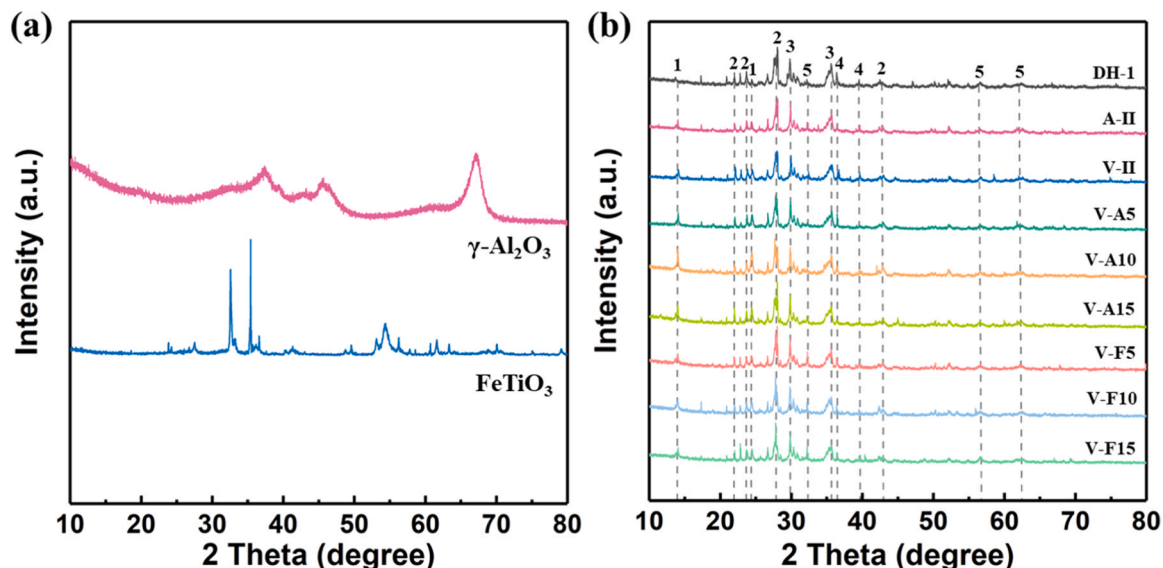


Fig. 6. XRD patterns of (a) γ -Al₂O₃ and FeTiO₃ additives and (b) the DH-1 based geopolymers. 1: zeolite; 2: plagioclase; 3: pyroxene; 4: olivine; 5: ilmenite.

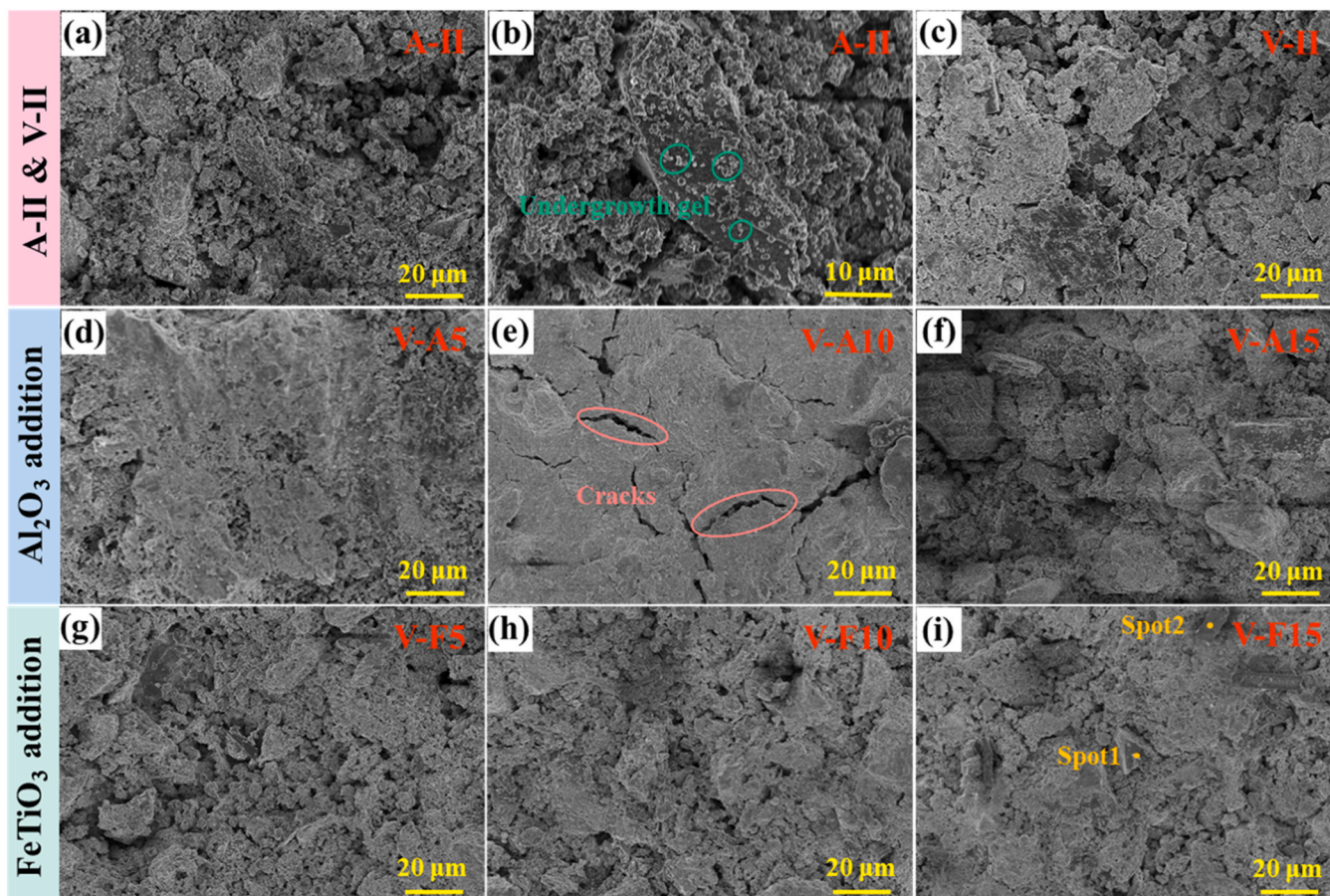


Fig. 7. SEM images of the DH-1 based geopolymers of (a) A-II, (b) A-II with high magnification, (c) V-II, (d) V-A5, (e) V-A10, (f) V-A15, (g) V-F5, (h) V-F10, and (i) V-F15.

composition of the flaky product. FeTiO_3 added geopolymers (Fig. 8g-i) demonstrate the product abundance comparable to V-II. Combined with XRD results, it can be concluded that the acicular features correspond to zeolite crystals, while the flaky morphology represents amorphous $\text{Al}(\text{OH})_3$.

Fig. 9 presents the elemental evolution characteristics of the DH-1 based geopolymers through EDS analysis. Comparative analysis reveals that the air-cured specimen A-II demonstrates a higher Si/Al ratio (3.15) than its vacuum-cured counterpart V-II (2.80). With increasing addition of $\gamma\text{-Al}_2\text{O}_3$, a systematic reduction in Si/Al ratios is observed, decreasing from 2.06 (V-A5) to 1.25 (V-A15). Note that optimal geopolymer network development occurs at the moderate Si/Al ratio, where controlled Al^{3+} substitution into silicate tetrahedra facilitates three-dimensional N-A-S-H gel formation through Si-O-Al crosslinking. In addition, A-II exhibits a lower Na/Si ratio (0.81) than V-II (0.87). $\gamma\text{-Al}_2\text{O}_3$ added specimens V-A5, V-A10, and V-A15 demonstrate increasing of Na/Si ratios to 0.94, 1.09, and 1.34, respectively. This indicates that Na is involved in the geopolymerization reaction, and its possible role is to neutralize the negative charge. In contrast, FeTiO_3 added specimens indicate elemental ratios comparable to V-II, maintaining stable Si/Al (~ 2.80) and Na/Si (~ 0.85) ratios, confirming limited chemical interaction between FeTiO_3 and the geopolymer matrix.

3.4. ^{27}Al MAS NMR analysis

^{27}Al NMR spectra (Fig. 10a) reveals two resonance peaks at ~ 10 ppm and ~ 60 ppm, corresponding to octahedral $[\text{AlO}_6]$ and tetrahedral $[\text{AlO}_4]$ coordination, respectively [39,40]. Quantitative results

(Fig. 10b) demonstrate that when the $\gamma\text{-Al}_2\text{O}_3$ addition up to 10 wt%, the relative area of $[\text{AlO}_4]$ species increases from 46.86 % to 50.37 %, accompanied by a corresponding elevation in the $[\text{AlO}_4]/[\text{AlO}_6]$ ratio from 0.88 to 1.06. This trend confirms more and more Al incorporation into the N-A-S-H gel network through Si-O-Al bridging. However, at 15 wt% Al_2O_3 addition, the $[\text{AlO}_4]$ fraction decreased to 44.03 %, and the coordination ratio dropped to 0.78. This inversion correlates with SEM-identified amorphous $\text{Al}(\text{OH})_3$ product in V-A15, wherein lots of Al^{3+} undergoes six-fold coordination via hydroxyl complexation rather than tetrahedral network integration.

3.5. FTIR analysis

Fig. 11 illustrates the FTIR spectra of the DH-1 based geopolymers, with band assignments detailed in Table 3. Notably in Fig. 11a, peak 1 at $\sim 475\text{ cm}^{-1}$ corresponds to the bending vibration of chemically inert Si-O-Si linkages [41]. Peak 2 at $\sim 580\text{ cm}^{-1}$ arises from symmetric stretching vibration of S-O-Al bond, indicative of crosslinking within the aluminosilicate framework. Peak 3 at $\sim 735\text{ cm}^{-1}$ is ascribed to the Si-O-Si or Si-O-Al symmetry stretching vibration [42]. The broad band 4 between 820 cm^{-1} and 1300 cm^{-1} is associated with the asymmetric stretching vibration of Si-O-T bonds, where T represents Si or Al [43,44]. A distinct carbonate-related peak 5 emerges at $\sim 1470\text{ cm}^{-1}$, assigned to O-C-O asymmetric stretching vibrations, likely from carbonation between atmospheric CO_2 and the alkaline activator [45,46]. Additionally, peak 6 and 7 at $\sim 1650\text{ cm}^{-1}$ and $\sim 3450\text{ cm}^{-1}$ are linked to H-O-H bond bending vibration and O-H bond stretching vibration, which derived from the water in air and additional water [47].

As shown in Fig. 11b, the centers of the broad peaks spanning

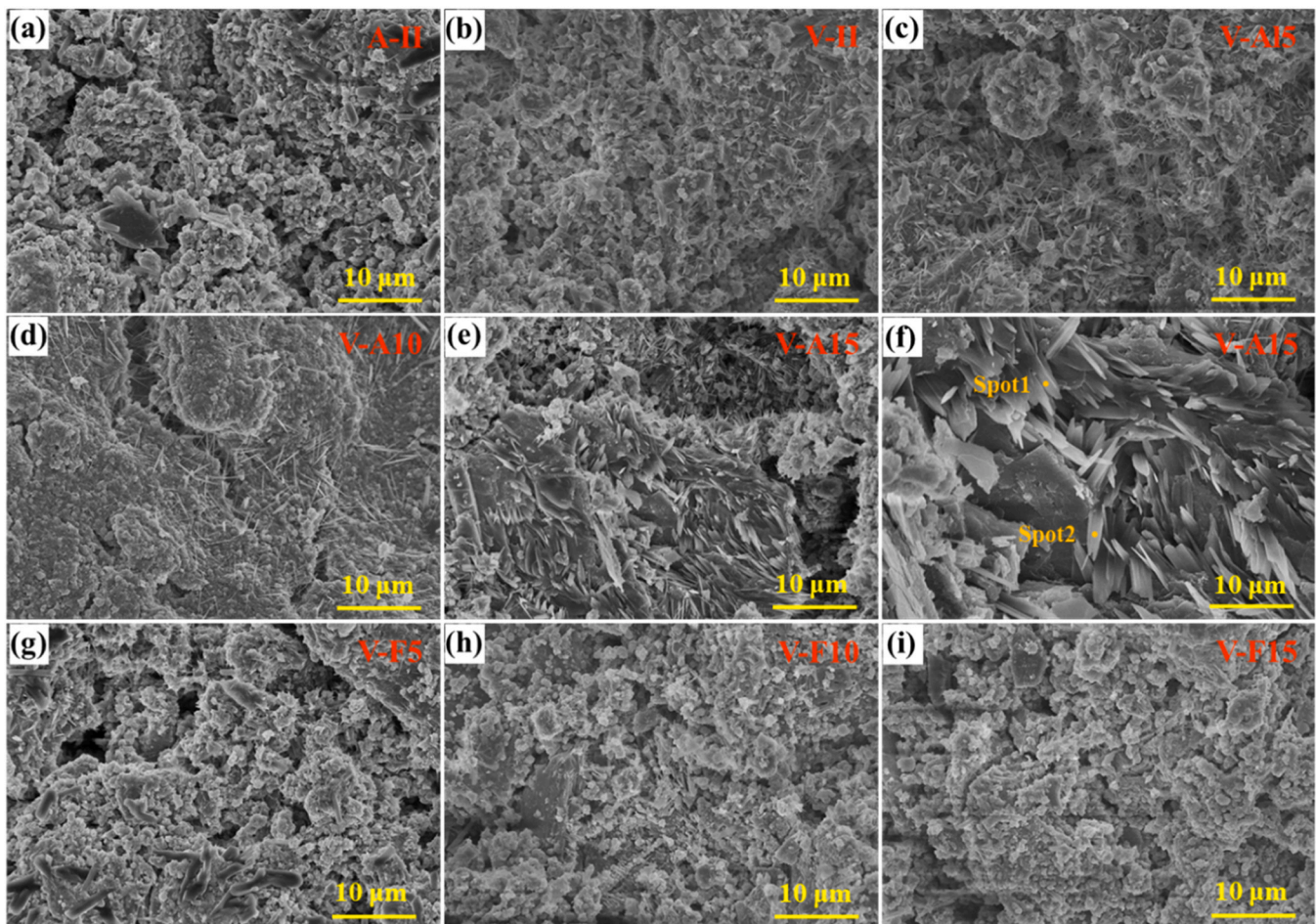


Fig. 8. High resolution SEM images of the DH-1 based geopolymers of (a) A-II, (b) V-II, (c) V-A5, (d) V-A10, (e) V-A15, (f) V-A15 with higher magnification, (g) V-F5, (h) V-F10, and (i) V-F15.

820–1300 cm^{-1} in specimens A-II and V-II shift to lower wavenumbers compared to DH-1, which is caused by the rearrangement of Si-O and Al-O [30]. The $\gamma\text{-Al}_2\text{O}_3$ added specimens (Fig. 11b) indicate a systematic shift pattern where the peak centers initially move to lower wavenumbers and then to higher wavenumbers, with V-A10 and V-A15 exhibiting extremal values at $\sim 993 \text{ cm}^{-1}$ and $\sim 1002 \text{ cm}^{-1}$, respectively. In contrast, FeTiO_3 added specimens (Fig. 11c) maintained spectral stability with peak centers consistently centered at $\sim 1000 \text{ cm}^{-1}$. In addition, the Si-O-Al bond at $\sim 580 \text{ cm}^{-1}$ and $\sim 735 \text{ cm}^{-1}$ display progressive intensity with $\gamma\text{-Al}_2\text{O}_3$ incorporation up to 10 wt% (Fig. 11b), indicating more original $[\text{SiO}_4]$ group on the uniform Si-O-Si chain structure in DH-1 is replaced by $[\text{AlO}_4]$ to form new amorphous gel. This is also proved by the SEM observations in Fig. 7. And no comparable intensity variation is observed in FeTiO_3 added specimens (Fig. 11c). Notably, the O-C-O stretching vibration at $\sim 1470 \text{ cm}^{-1}$ manifests the highest intensity in air-cured specimen A-II among DH-1 based geopolymers.

The spectral region between 820 and 1300 cm^{-1} is deconvoluted to quantify amorphous phase constituents (Fig. 12). The sub-band at $\sim 975 \text{ cm}^{-1}$ arise from the asymmetric tensile vibration of Si-O-T bonds in the N-A-S-H gel [48–50]. Quantitative analysis (Fig. 12i) reveals distinct curing-dependent variations. The air-cured specimen A-II exhibits the lowest area (6.7 %), compared to 8.8 % for its vacuum-cured counterpart V-II. In addition, Gel content increased non-linearly with $\gamma\text{-Al}_2\text{O}_3$ addition, reaching a maximum (12.4 %) in V-A10, then dropping to 7.1 % in V-A15, indicating an optimal Al incorporation threshold. However, FeTiO_3 added systems maintain stable areas within a narrow range (8.8–9.0 %), indicating negligible influence on

geopolymerization.

The gel content trends from FTIR are supported by complementary techniques. SEM images (Fig. 7) show a progression of microstructural development that aligns with the FTIR results. The systematic evolution of the Si/Al ratio measured by EDS (Fig. 9i) provide consistent evidence for the gel content increase inferred from FTIR. Furthermore, the increase in tetrahedrally coordinated $[\text{AlO}_4]$ quantified by ^{27}Al MAS NMR (Fig. 10) validates the enhanced geopolymerization upon moderate $\gamma\text{-Al}_2\text{O}_3$ addition. Conversely, the precipitous drop in gel content for V-A15 is corroborated by the NMR-observed shift to octahedral $[\text{AlO}_6]$ and the SEM-observed formation of amorphous $\text{Al}(\text{OH})_3$.

To quantitatively assess the role of the N-A-S-H gel, the relative gel area from FTIR is plotted against 3-day compressive strength (Fig. S5). A strong positive correlation is observed for most specimens, confirming that the gel content is the primary determinant of mechanical performance. The significant deviation of V-A10 from this trend—exhibiting the highest gel content but a sub-optimal strength, provides quantitative evidence that its mechanical properties are compromised by the microcracking observed in SEM (Fig. 7e), highlighting the critical influence of microstructural integrity beyond mere gel content.

3.6. Mechanism analysis

Fig. 13 illustrates the geopolymerization reaction mechanism. Under highly alkaline conditions, aluminosilicate precursors undergo surface activation through cleavage of Si-O-T bonds (where T = Si or Al), resulting in the dissolution of $[\text{SiO}_4]$ and $[\text{AlO}_4]$ tetrahedral species into the aqueous phase. Subsequently, Na^+ reacts with these $[\text{SiO}_4]$ and

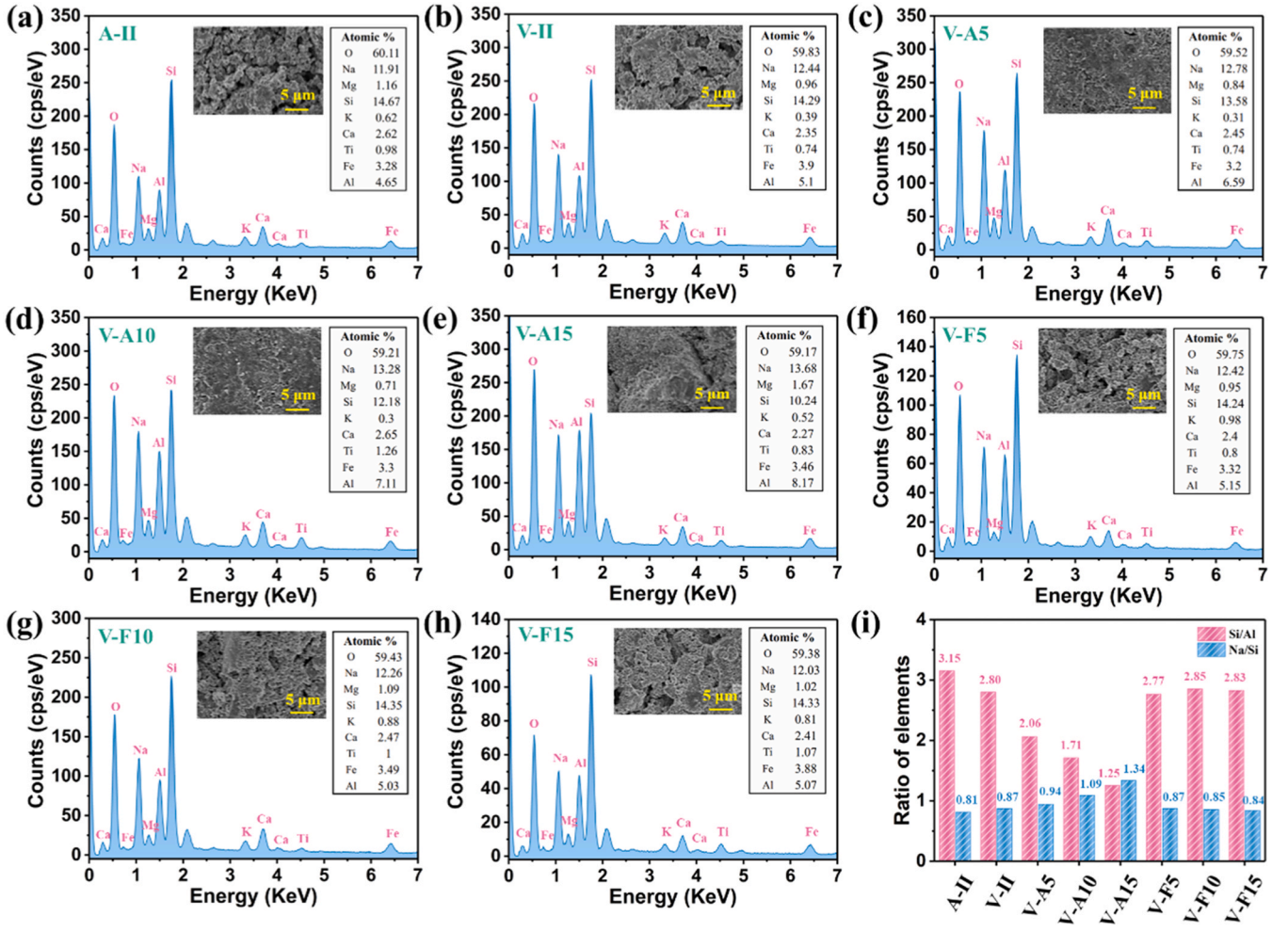


Fig. 9. SEM images and EDS analysis of the DH-1 based geopolymers of (a) A-II, (b) V-II, (c) V-A5, (d) V-A10, (e) V-A15, (f) V-F5, (g) V-F10, and (h) V-F15. (i) The Si/Al and Na/Si ratios detected by EDS tests.

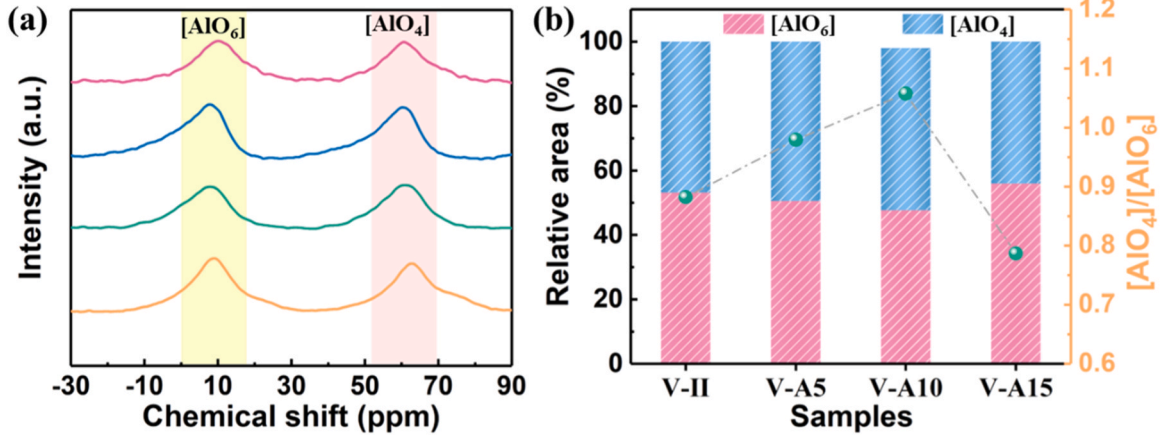


Fig. 10. (a) ^{27}Al MAS NMR spectra and (b) the relative areas of the $[\text{AlO}_4]$ and $[\text{AlO}_6]$ species of V-II and $\gamma\text{-Al}_2\text{O}_3$ addition group.

$[\text{AlO}_4]$ units in order to realize the charge balance of the system and combines with a portion of water to form a $[-\text{Si-O-Al-}]_n$ gel network with a three-dimensional structure [51,52]. Obviously, the amount of N-A-S-H gel is critical for compressive strength. Complementary factors influencing mechanical performance are systematically investigated through DH-1 based geopolymers, as detailed in Fig. 14.

Comparative analysis demonstrates a significant 38.5 % reduction in

strength for the air-cured specimen A-II (5.9 ± 0.8 MPa) relative to its vacuum-cured counterpart V-II (9.6 ± 0.3 MPa). This performance disparity is attributed to two synergistic mechanisms. First, carbonation occurs as atmospheric CO_2 reacts with dissolved NaOH, forming poorly soluble Na_2CO_3 . This reduces the effective concentration of the alkaline activator available for geopolymerization. Second, air curing inevitably entrains numerous microbubbles within the particle matrix and

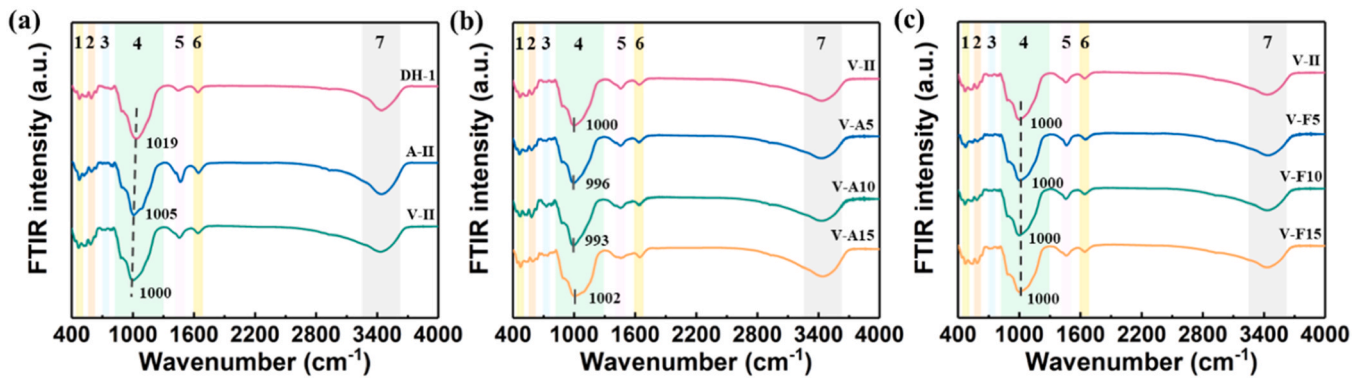


Fig. 11. FTIR spectra of the DH-1 based geopolymers: (a) DH-1, A-II, and V-II, (b) V-II and γ - Al_2O_3 additives group, and (c) V-II and FeTiO_3 additives group.

Table 3

FTIR species of the DH-1 based geopolymers remarked in Fig. 11.

Band	Wavenumber (cm^{-1})	Species
1	475	Si-O-Si bending
2	580	Si-O-Al symmetric stretching
3	735	Si-O-Si or Si-O-Al symmetric stretching
4	820–1300	Si-O-T (T = Si/Al) asymmetric stretching
5	1470	O-C-O stretching
6	1650	H-O-H bending
7	3450	-OH stretching

interstitial spaces, even post-vibration, whereas vacuum processing achieves near-complete bubble elimination. These bubbles reduce the interfacial contact area between reactant particles. Consequently, A-II undergoes insufficient geopolymerization, yielding less N-A-S-H gel and inferior strength. Furthermore, the low solubility of Na_2CO_3 induces its precipitation and subsequent volumetric expansion, generating additional defects (see Fig. S6) that further decrease the compressive strength [34].

In γ - Al_2O_3 added systems, specimen V-A5 achieves maximum strength (13.1 ± 0.7 MPa) through optimal crosslinking of N-A-S-H gel networks interspersed with short acicular zeolite crystals. In contrast, specimen V-A10 (8.8 ± 0.5 MPa) exhibits strength reduction despite possessing the highest N-A-S-H gel content (12.4 %, Fig. 12i). This paradox is primarily attributed to the synergistic effect of charge imbalance-induced stress and the formation of a brittle microstructure. The high Al incorporation leads to a highly cross-linked but brittle network. More critically, the excessive $[\text{AlO}_4]$ likely exceed the local charge-balancing capacity of Na^+ cations, creating localized charge disparities. This imbalance generates electrostatic repulsion within the gel matrix, introducing internal tensile stresses. Coupled with the enhanced network rigidity and drying shrinkage, these internal stresses promote microcrack propagation (Fig. 7e), which degrades the compressive strength despite the high gel volume.

For V-A15, the severe strength loss (2.6 ± 0.1 MPa) originates from a fundamental shift in the reaction pathway. The massive surplus of Al^{3+} from 15 wt% γ - Al_2O_3 addition consumes the alkaline activator, depleting it for its primary role of dissolving silicate species and maintaining high pH for geopolymerization. Consequently, a significant portion of the Al^{3+} precipitates as amorphous $\text{Al}(\text{OH})_3$, a mechanically weak and non-cementing phase, as identified by EDS (Fig. S4) and its flaky morphology (Fig. 8f). This precipitation reaction permanently removes active Al^{3+} from the geopolymerization process, as confirmed by the ^{27}Al NMR results (Fig. 10), which shows a drastic decrease in tetrahedral $[\text{AlO}_4]$ (the desired form in N-A-S-H gel) and a corresponding dominance of octahedral $[\text{AlO}_6]$ (characteristic of $\text{Al}(\text{OH})_3$) in V-A15. Consequently, the gel content quantified by FTIR plummets to 7.1 % (Fig. 12i), the lowest among all modified specimens, directly explaining the loss of structural integrity and strength.

FeTiO_3 added specimens exhibit comparable amorphous gel and crystal content to control V-II, with chemical inertness in alkaline media confirmed by XRD and FTIR. Nevertheless, V-F15 demonstrates modest strength enhancement (10.3 ± 0.4 MPa vs. 9.6 ± 0.3 MPa) through particle-induced pore-filling densification (see Fig. 7i).

4. Conclusion

This study successfully synthesis of DH-1 LRS, which closely matches the chemical and mineralogical properties of CE-5 LR, and converted it into geopolymers via NaOH activation. Systematic investigation of curing parameters and supplementary additives yielded the following principal conclusions:

- (1) Compressive strength development strongly depends on curing temperature. While elevated average curing temperature over 72 h enhances geopolymerization, early-stage curing temperature (within the initial 24 h) exerts a dominant influence on final mechanical performance.
- (2) Vacuum-cured specimens exhibit a 62.7 % higher compressive strength than air-cured counterparts. This performance superiority is attributed to the elimination of atmospheric CO_2 induced carbonation, preventing Na_2CO_3 formation, associated alkaline activator consumption, and deleterious volumetric expansion defects. And effective degassing of the slurry, maximizing reactant particle contact area and promoting complete N-A-S-H gel network formation.
- (3) Intermediate γ - Al_2O_3 addition (5 wt%, specimen V-A5) maximizes compressive strength (13.1 ± 0.7 MPa) by optimizing N-A-S-H gel crosslinking density and microstructure. Higher additions induce detrimental effects. At 10 wt% (V-A10), excessive network rigidity and localized charge imbalance ($[\text{AlO}_4] > \text{available Na}^+$) promote crack propagation despite maximal gel content. At 15 wt% (V-A15), precipitation of amorphous $\text{Al}(\text{OH})_3$ consumes active Al^{3+} and alkaline activator, drastically impeding gel formation and reducing strength by 80.2 % relative to V-A5.
- (4) FeTiO_3 incorporation had negligible chemical interaction with the geopolymer matrix, as confirmed by invariant elemental ratios (Si/Al, Na/Si), phase evolution (XRD), and gel content (SEM, FTIR). Modest strength gains observed at higher loadings (10.3 ± 0.4 MPa for V-F15) are solely attributable to physical pore-filling densification by inert particulates.

Despite promising mechanical performance at the laboratory scale, practical deployment of LR-based geopolymers faces significant scalability challenges related to resource availability and engineering implementation. First, reliance on water as a reaction medium is a critical constraint given the Moon's extreme water scarcity. Future work should focus on developing low-water or anhydrous geopolymerization

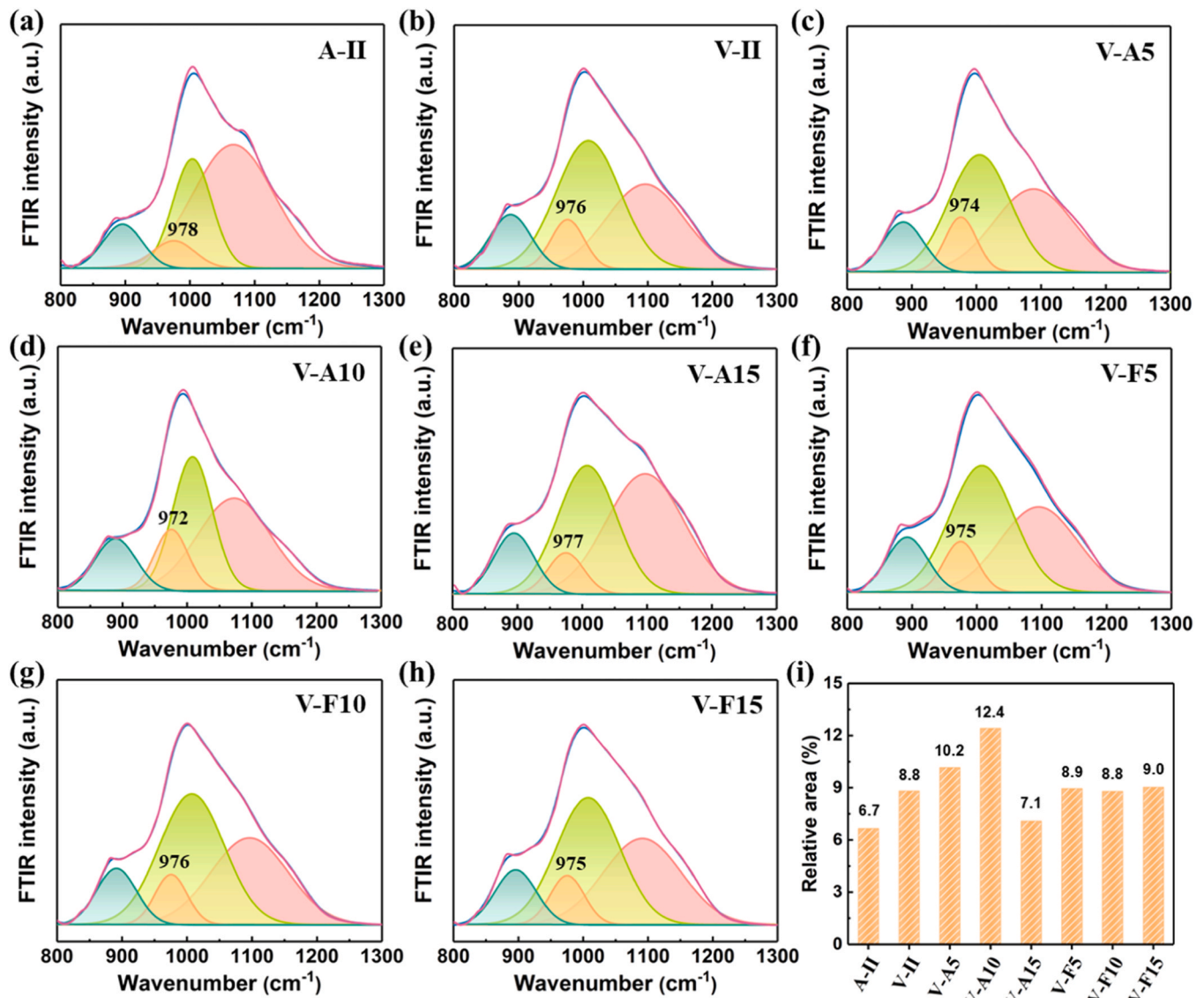


Fig. 12. (a-h) FTIR spectra peak separation results of the DH-1 based geopolymers between 800 and 1300 cm^{-1} and (i) the relative area of the bonds representing N-A-S-H gel.

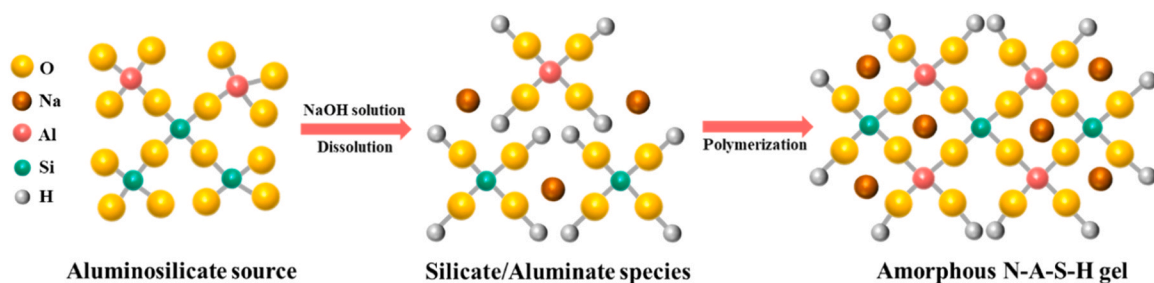


Fig. 13. Schematic diagram of the reaction mechanism of geopolymerization.

systems. One potential water source is extraction from lunar regolith via thermal processing: solar wind-implanted hydrogen can undergo reduction reactions with iron oxides at high temperatures, producing metallic iron and water vapor. Second, while vacuum curing proved beneficial in this study, maintaining a large-scale preparation on the moon is energy-intensive and engineeringly complex. Addressing these challenges will require careful selection of LR raw materials with

suitable particle sizes and chemical composition, along with the development of intelligent, automated additive manufacturing systems tailored to in-situ conditions.

Additionally, the long-term performance and durability of LR based geopolymers under lunar conditions require further research. Key aspects include evaluating their resistance to high-energy ultraviolet radiation and galactic cosmic rays, which may induce structural

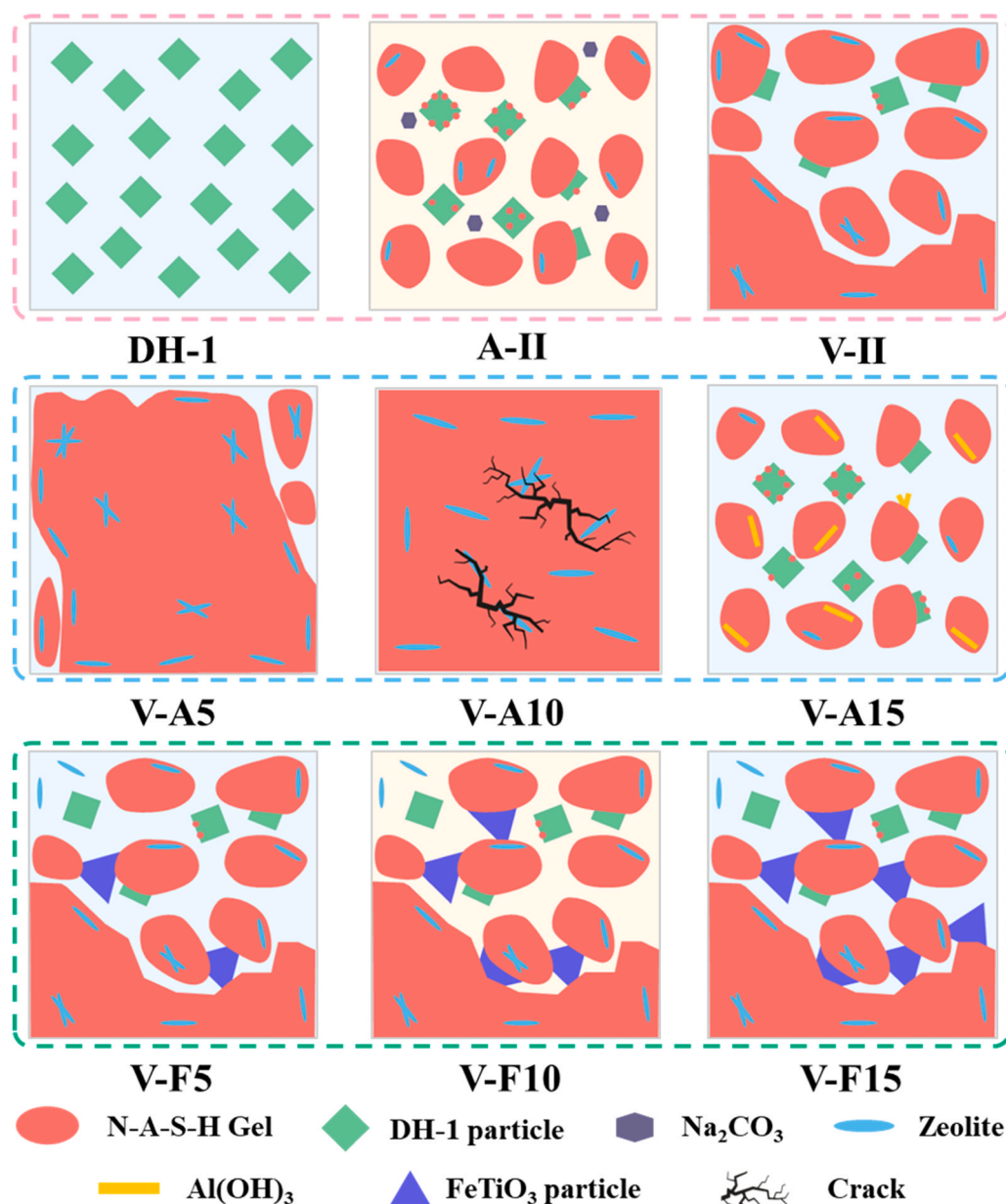


Fig. 14. Schematic diagram of the microstructural evolution of the DH-1 based geopolymers.

modifications in the aluminosilicate network. Resistance to shock waves and thermal fatigue from micrometeoroid impacts also needs evaluation. Furthermore, while this study clearly shows how mix design affects compressive strength, developing a generalized predictive model for lunar geopolymers requires a broader dataset covering more mixture proportions and curing conditions. Future work will focus on generating such a dataset to enable the application of advanced machine learning techniques with rigorous training/validation/testing protocols, thereby creating a powerful tool for optimizing lunar construction materials. This study provides a fundamental understanding of geopolymer behavior under lunar-like environments, overcoming these challenges is the essential next step towards making lunar geopolymerization a viable ISRU technology.

CRediT authorship contribution statement

Yanjie Shi: Investigation, Formal analysis. **Mengzhao Ding:** Writing – review & editing, Writing – original draft, Methodology, Investigation,

Data curation, Conceptualization. **Qingwei Wang:** Resources, Project administration, Conceptualization. **Lida Luo:** Supervision, Resources. **Huimin Han:** Investigation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supporting information

Supplementary data associated with this article can be found in the

online version at doi:10.1016/j.conbuildmat.2025.143672.

Data availability

Data will be made available on request.

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